SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1717**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I Krisha Meenatchi M (192511338) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Krisha Meenatchi M

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

<u>RUBRICS FOR CALCULATION:</u>

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article’s purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Task 1: Article Summary & Citation Details

(a) Article Citation & Domain Parameters

Full Citation:
Judy Simon, Nellore Kapileswar, and Anoop Mohanakumar, "A Novel Energy Adaptive Neural Network and Deep Q-Learning Network for Improved Energy Efficiency in Dynamic Underwater IoT Environment," IEEE Access, vol. 13, pp. 1-14, 2025. DOI: 10.1109/ACCESS.2025.3597025.

Domain / Application Area:
Underwater Internet of Things (UIoT) networks utilizing a hybrid optical-acoustic wireless communication system. Practical deployment areas include marine environmental monitoring, offshore oil pipeline inspection, and autonomous underwater vehicle (AUV) coordination.

Specific Machine Learning Problem Addressed:
Joint optimization of transmission power control, frequency selection, dynamic modulation, and adaptive error correction. The models must adapt to highly dynamic, unpredictable underwater channel conditions (high propagation delays, water turbidity, scattering, multipath fading, and variable ambient noise) to achieve a multi-objective goal: minimizing sensor node energy consumption while maintaining a high communication quality threshold.

(b) Key Research Objective & Identified Gaps

Research Gaps:
The authors identified that conventional underwater routing protocols (e.g., LEACH, PEGASIS, TEEN, and MTE) are static and fail to adapt to severe underwater channel variations. Although some hybrid optical-acoustic models have been proposed to combine high-speed, short-range optical links with robust, long-range acoustic links, they suffer from critical limitations: (1) power management and modulation schemes are typically designed independently, ignoring their deep physical interdependence; (2) existing deep learning strategies focus primarily on routing or power alone, leaving modulation configuration static; and (3) prior models ignore the operational latencies and energy overheads of dynamic channel-switching, resulting in unrealistic performance evaluations in fluctuating marine environments.

Proposed Solution & Objectives:
To address these gaps, the authors proposed a unified, deep learning-driven framework that simultaneously optimizes power control and modulation reliability. The core objective is realized through a dual-model integration: an Energy-Adaptive Neural Network (EANN) that dynamically scales transmission power and selects frequency based on real-time Channel State Information (CSI), and a Deep Q-Learning Network for Adaptive Error Correction and Modulation (DQL-AECM) that adjusts the error correction codes (ECC) and modulation schemes based on instantaneous Signal-to-Noise Ratio (SNR). This joint strategy minimizes unnecessary energy consumption while ensuring the system maintains a robust, reliable communication link (above a specified minimum quality, Q_{min}) even during severe environmental transitions.

Task 2: Methodology Analysis & Syllabus Mapping

(a) Machine Learning Techniques & Simulation Dataset Details

1. Energy-Adaptive Neural Network (EANN) for Power and Frequency Control:

• **Architecture & Hyperparameters:** The EANN is structured as a 4-layer feedforward neural network, consisting of one input layer, two fully connected hidden layers with 128 neurons each, and one output layer. The hidden layers apply the non-linear Rectified Linear Unit (ReLU) activation function to accelerate convergence. The output layer utilizes a linear activation function to predict continuous real values representing the recommended transmission power $P(t)$ and frequency $f(t)$.

• **Inputs & Mathematical Formulation:** The input vector at each time step t is $x(t) = [h(t), P(t-1), f(t-1), E(t-1), Q(t-1)]$, containing Channel State Information $h(t)$ (modeling path loss, multipath fading, noise, and attenuation), previous power, previous frequency, energy consumption, and communication quality. The objective is to minimize total energy consumption $E(t) = P(t) * T(t)$ subject to keeping the estimated communication quality $Q(h(t), f(t)) >= Q_{min}$.

• **Optimization:** Gradient descent is applied iteratively using a composite loss function to update the parameters, weighted by alpha = 0.7 for energy efficiency and beta = 0.3 for quality.

2. Deep Q-Learning for Adaptive Error Correction and Modulation (DQL-AECM):

• **Reinforcement Learning Structure:** This model dynamically adjusts error correction codes and modulation schemes in response to fluctuating SNR. The State Space $s(t)$ is defined as $[gamma(t), ER(t), M(t), C(t)]$ representing SNR, current bit error rate, modulation scheme, and error correction code. The Action Space $a(t)$ comprises discrete combinations of available modulation schemes and error correction codes. The Reward Function is formulated to penalize both error rate and energy usage: $R(s(t), a(t)) = -(lambda_1 * ER(t+1) + lambda_2 * E(t+1))$.

• **Deep Q-Network & Experience Replay:** The Q-function is approximated using a deep neural network with an input layer, three hidden layers (128, 64, and 32 neurons with ReLU activation), and an output layer with a linear activation to output unbounded Q-values. The model utilizes an epsilon-greedy policy for the exploration-exploitation trade-off and implements experience replay with a replay buffer of 50,000 experiences, a batch size of 32, and a target network update frequency of 500 steps.

3. Simulation Dataset & Preparation:

• **Simulation Setup:** Because real-world physical datasets for underwater hybrid links are extremely difficult to harvest, the dataset was generated using the specialized Aqua-Sim simulation environment. The simulation network comprises 100 static nodes randomly deployed over a 3D coordinates volume of 100 * 100 * 50 meters, communicating for 2000 seconds.

• **Features & Environment Noise:** Input features represent time-varying underwater physical parameters: distance between nodes, attenuation coefficients, path loss, and dynamic ambient noise. The noise power spectral density was modeled between -120 to -90 dBm to account for surface turbulence, marine life, and background acoustics. Highly reliable, short-range optical channels experience noise closer to -120 dBm (operating at <= 30 meters), whereas acoustic channels experience noise between -115 dBm and -90 dBm.

• **Validation:** To ensure statistical validity, 20 independent simulation trials were executed, and the standard deviation of key performance metrics remained within a highly stable +/-5% margin.

(b) Syllabus Mapping, Strengths, and Limitations

Syllabus Mapping (CO 4 Connection):
This research directly aligns with the core components of Course Outcome 4 (CO4). The DQL-AECM model is a direct application of Reinforcement Learning, implementing key curriculum concepts such as Markov Decision Processes (MDP), state-action-reward representations, temporal difference learning, and neural networks as function approximators (Deep Q-Networks). The EANN model represents building models for classification and decision-making, where deep feedforward architectures are trained to make real-time multi-objective regression decisions (transmission power and frequency selection) under uncertainty.

Methodological Strength:
• **Joint Integration and Realistic Switching:** The framework's primary strength is its unified structure. Rather than isolating power and modulation, it treats them as interdependent variables. Furthermore, the simulation explicitly incorporates a transition latency of 1.5 seconds and an average energy penalty of 0.3 Joules per mode switch between optical and acoustic links. This modeling of switching overhead prevents the over-optimism common in simulated networks, making the evaluated performance metrics highly rigorous and physically meaningful.

Methodological Limitation:
• **Computational Complexity and Centralization:** The forward pass complexity of EANN is $O(L * n^2 + f * n)$, and DQL-AECM step complexity is $O(d * k^2 + |S| * |A|)$. Training requires standard GPU acceleration (NVIDIA RTX 3080). This presents a critical limitation: underwater sensor nodes are tiny, low-power edge microcontrollers with extremely restricted battery capacities. Running complex neural network updates in real-time could drain more computational energy than the physical transmission energy saved. Additionally, the single-agent learning framework lacks decentralized multi-agent coordination, restricting the protocol's scalability in larger or dynamically mobile deployments.

Task 3: Results & Critical Evaluation

(a) Benchmarking Analysis & Comparative Performance Table

The proposed dual-model framework was benchmarked against four traditional routing and energy protocols adapted to the underwater environment: LEACH, PEGASIS, TEEN, and MTE (Minimum Transmission Energy). The comparative metrics below summarize the performance over a 2000-second simulation cycle:

Protocol Model	Total Energy (Joules @ 2000s)	Energy per Bit (10^-6 J/bit)	Steady SNR (dB @ 2000s)	Steady-State BER	Max Throughput (Mbps @ 100s)	Latency (ms @ 100s)
Proposed Model	**22.0 J**	**2.0**	**21.0 dB**	**10^-8**	**6.2 Mbps**	**35 ms**
MTE Model	28.0 J	2.6 (est)	16.0 dB	< 10^-2	2.5 Mbps	60 ms

PEGASIS Model	33.0 J	2.8 (est)	19.0 dB	10^-4	3.5 Mbps	52 ms
LEACH Model	35.0 J	3.4	18.0 dB	10^-2	4.2 Mbps	45 ms
TEEN Model	37.0 J	3.7	17.0 dB	< 10^-2	3.0 Mbps	55 ms

> **Key Performance Summary:** *The proposed deep learning framework significantly outperforms traditional algorithms in both efficiency and reliability. At the end of 2000 seconds, the Proposed Model consumes only 22 Joules of energy, which is 21.4% less than MTE (28 J) and 40.5% less than TEEN (37 J). Concurrently, the Proposed Model establishes a highly robust communication link, stabilizing at a Bit Error Rate (BER) of 10^-8 and a Signal-to-Noise Ratio (SNR) of 21 dB, whereas the next best performing baseline model (PEGASIS) achieves a BER of only 10^-4 at 19 dB, and LEACH drops to a high-error rate of 10^-2. This demonstrates that the dynamic learning-driven joint optimization effectively eliminates the trade-off bottleneck between power savings and link reliability.*

(b) Critical Evaluation of Metrics, Biases, and Gaps

Are the Reported Metrics Realistic?
The metrics are supported by strong internal validation. First, the authors conducted 20 independent simulation trials to calculate standard deviation, proving that the standard deviation of SNR, BER, and throughput remains within +/-5% across the -120 to -90 dBm noise spectrum. Second, a thorough sensitivity analysis of loss function weights (alpha and beta) validates the EANN's design: when alpha is too high (>0.8, representing extreme energy focus), communication degrades with BER rising beyond 10^-4; when alpha is too low, energy spikes to 29 Joules. Setting alpha = 0.7 and beta = 0.3 demonstrates a realistic and optimized operating trade-off.

Potential Evaluation and Dataset Biases:
• **Simulation Environment Bias:** The entire framework was evaluated within the Aqua-Sim simulation tool with static sensor nodes. Simulated ocean conditions are highly idealized and do not model dynamic complex physical channel features such as continuous ocean currents, marine tidal drifts, biofouling on optical lenses (which rapidly degrades short-range optical communication), salinity fluctuations, or temperature stratification. These omissions can make simulated links appear far more stable than physical underwater channels.
• **Computation/Hardware Evaluation Bias:** The model's training and execution times (<2.3 seconds per EANN cycle and <1.8 seconds per DQL iteration) are measured on an NVIDIA RTX 3080 GPU setup. Real-world underwater sensor nodes are deployed on tiny, battery-powered edge microcontrollers that completely lack GPU hardware. The paper presents a computational bias by assuming the algorithm's real-time feasibility without testing on representative, low-power embedded processors.

Suggested Experiments to Strengthen the Findings:
1. Hardware-in-the-Loop (HIL) Testing: Deploy the EANN and DQL-AECM networks onto a physical low-power microcontroller (such as an ARM Cortex-M4 or a specialized low-power DSP) to record the exact computational execution delay and processing energy draw.

2. Mobility and Spatial Drift Simulation: Introduce node mobility in Aqua-Sim to model actual ocean current drift, assessing how the algorithms handle rapid spatial and distance-based topology changes and intermittent link disconnections.

(c) Industrial Applicability & Scaling Challenges

Practical Deployment Opportunities:
The dual-model framework is highly applicable to several mission-critical maritime sectors: (1) Marine Environmental Monitoring to track long-term sea-temperature, pH levels, and marine life; (2) Offshore Oil & Gas Pipeline Inspection where deep-water pipeline integrity can be mapped in real-time by coordination between sensor arrays and Autonomous Underwater Vehicles (AUVs); and (3) Strategic Ocean Defense and mapping swarm operations.

Key Challenges in Scaling to Industry:
• **Extreme Edge Compute Limitations:** The O(L * n^2) complexity of the feedforward passes and the DQL update loops represent significant computational burdens. Underwater nodes must operate autonomously for several years on tiny batteries. Without model optimization, the energy consumed by the edge processor executing neural network iterations can exceed the transmission energy saved, rendering the system counterproductive.
• **Site-Specific Channel Characteristics:** Deep learning models require highly reliable, localized datasets. Underwater environments are extremely site-specific—coastal water turbidity differs dramatically from the deep, clear ocean. Collecting and preparing localized CSI training datasets for every new deployment is a highly complex, labor-intensive industrial barrier.
• **Ecological Sonar Footprint:** Deploying massive, industrial-scale underwater sensor networks that continuously transmit acoustic signals can disrupt marine ecosystems. Marine mammals (such as whales, dolphins, and seals) rely on sonar frequencies for navigation, communication, and foraging. Widespread acoustic transmissions present potential ecological risks that must be analyzed in accordance with local maritime environmental protection laws.

Task 4: Reflection & Learning Outcome

(a) Core Insights & Syllabus Connection

Insight 1: Multi-Objective Reinforcement Learning (MORL) & Reward Shaping:
• **Concept:** The DQL-AECM model utilizes a reward function R = -(lambda_1 * ER + lambda_2 * E) to penalize both bit error rate (reliability) and energy consumption (efficiency).
• **Syllabus Connection:** This directly maps to the Reinforcement Learning syllabus under CO4, demonstrating how reward shaping is critical to guide an RL agent toward a balanced, Pareto-optimal policy when optimizing competing variables in real-time.

Insight 2: Deep Q-Networks (DQN) as Non-Linear Function Approximators:
• **Concept:** The model maps a high-dimensional continuous state space (SNR, error rate, existing modulation, and coding) to discrete actions by utilizing a deep feedforward network to output Q-values instead of using a traditional lookup table.
• **Syllabus Connection:** This reinforces the transition from tabular Q-Learning to Deep Q-Networks (DQN) studied under Decision-Making algorithms in CO4, highlighting how deep neural networks act as

robust function approximators to resolve the "curse of dimensionality" in real-world environments.

Insight 3: Sensitivity of Loss Function Formulation:
• **Concept:** The EANN sensitivity analysis shows that minor adjustments in loss weights (alpha and beta) result in extreme behavioral shifts, where over-emphasizing energy (alpha > 0.8) completely destroys reliability.
• **Syllabus Connection:** This maps directly to the Statistical Learning and Model Building topics in CO4, illustrating that neural networks are not black-box models; rather, their performance, convergence, and operational trade-offs are strictly governed by mathematical loss formulation and hyperparameter calibration.

(b) Proposing a Novel Enhancement: Edge-Optimized Multi-Agent Coordination

Enhancement Proposal:
I propose a novel framework that combines **Multi-Agent Reinforcement Learning (MARL)** with **Post-Training Quantization (PTQ) and Neural Network Pruning**, deployed on a dynamic swarm of mobile Autonomous Underwater Vehicles (AUVs).

Justification of Feasibility:
• **Network Compression (PTQ & Pruning):** Pruning reduces the network size by removing redundant connections, while PTQ converts the model's weights from 32-bit floating-point format to 8-bit integers. This reduces memory footprint by approximately 75% and accelerates inference speed by 5x to 10x with less than 1% degradation in accuracy. This enables seamless, real-time local execution on standard ultra-low-power ARM Cortex-M microcontrollers.
• **Decentralized Coordination (MARL):** Using decentralized MARL algorithms (such as MAPPO) allows individual underwater nodes to act as independent agents that coordinate communication power and routing dynamically with nearby neighbors, removing the need for a high-overhead centralized gateway.

Expected Impact:
• **Real-Time Edge Viability:** Lowers computational energy drain below the transmission savings, achieving a true net-positive energy balance for battery-powered sensor nodes.
• **Scalability:** Enables scaling the network to thousands of active, mobile underwater nodes without a routing bottleneck, expanding applicability to broad oceanographic research and commercial offshore pipeline networks.
• **Mobility-Aware Optimization:** Ensures stable communication under rapid spatial changes, minimizing packet loss and optimizing throughput during marine drift.